CHEMICAL REACTION NETWORKS
1. Deterministic modelling

The *Law of Mass Action* and models based on Ordinary Differential Equations (ODEs)

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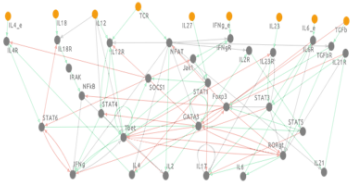
Cell Pathways and Chemical Reaction Networks (CRNs)

Biological networks

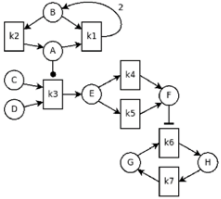
Cellular processes can be described in terms of networks, such as:

Gene regulatory networks



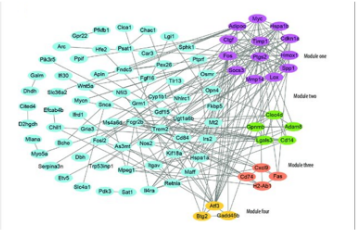
Describe influence relationships (activation/inhibition) between genes

Cell pathways



Describe chemical reactions in cells involving proteins, and other molecules

Protein-protein interaction networks



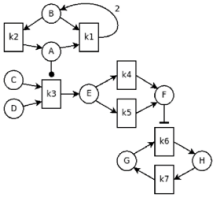
Describe interactions between proteins

Biological networks

Gene regulatory networks

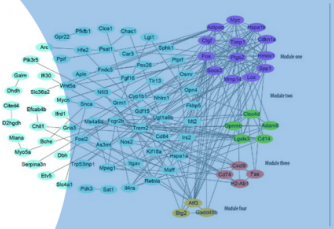
Describe influence relationships (activation/inhibition) between genes

Cell pathways



Describe chemical reactions in cells involving proteins, and other molecules

Protein-protein interaction networks



Describe interactions between proteins

We start with CHEMICAL REACTION NETWORKS (CRNs) since they provide the most accurate representation of a biological system dynamics

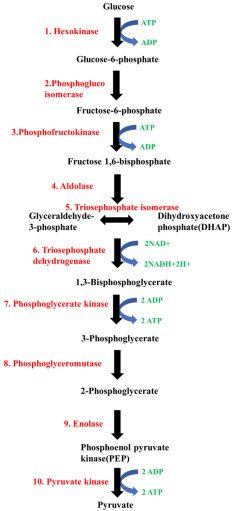
- A Cell Pathway is a CRN describing a specific process taking place in cells.

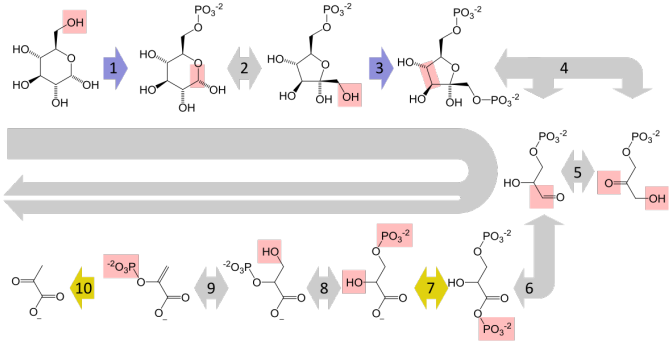
Examples of Cell Pathways: the Glycolysis (metabolic) pathway

Glycolysis: Overview

- The first metabolic pathway in cellular respiration.
- Occurs in the cytoplasm of all cells.
- Converts one molecule of glucose into two molecules of pyruvate.
- Generates a net gain of 2 ATP and 2 NADH.
- Functions under aerobic and anaerobic conditions.

Examples of Cell Pathways: the Glycolysis (metabolic) pathway





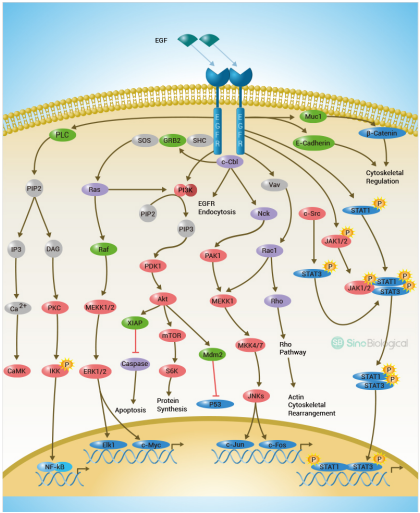
Examples of Cell Pathways: the EGF (signaling) pathway

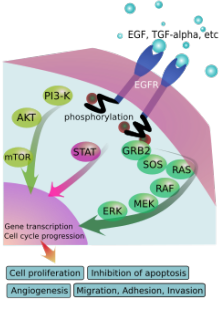
EGF Signaling Pathway: Overview

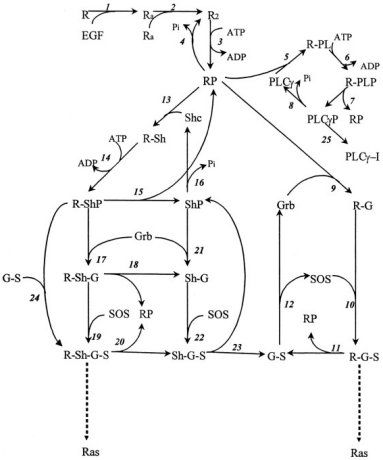
- The Epidermal Growth Factor (EGF) pathway regulates cell growth, proliferation, and differentiation.
- Begins with EGF binding to its receptor (EGFR), a transmembrane tyrosine kinase.
- Receptor dimerization triggers autophosphorylation of intracellular domains.
- Activates downstream cascades, e.g. RAS–RAF–MEK–ERK and PI3K–AKT pathways.
- Leads to transcriptional changes that control cell cycle progression and survival.
- Dysregulation of this pathway is often linked to cancer development.

A key model for studying signal transduction and network dynamics in systems biology.

Examples of Cell Pathways: the EGF (signaling) pathway



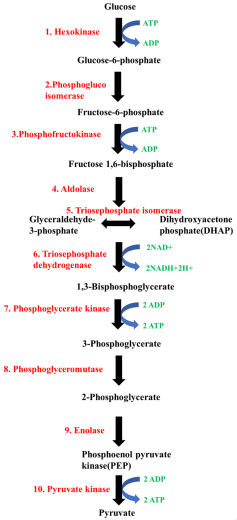




Pathways are Chemical Reaction Networks (CRNs)

- Biological pathways can be viewed as collections of chemical reactions.
- Each reaction transforms specific reactants into products.
- Reactions within a pathway often share common species
  - common reactant or products
  - a product of one may serve as a reactant for another
- This interconnection creates a network structure: the Chemical Reaction Network (CRN).
- CRNs capture the dynamic behavior of biochemical systems:
  - how molecules are transformed over time
  - fundamental for systems biology modelling

Glycolysis as a Set of Chemical Reactions



$$\text{Glucose} + \text{ATP} \longrightarrow \text{Glucose}-6\text{-phosphate} + \text{ADP}$$

$$\text{Glucose}-6\text{-phosphate} \rightleftharpoons \text{Fructose}-6\text{-phosphate}$$

$$\text{Fructose}-6\text{-phosphate} + \text{ATP} \longrightarrow \text{Fructose}-1{,}6-\text{bisphosphate} + \text{ADP}$$

$$\text{Fructose}-1{,}6-\text{bisphosphate} \longrightarrow \text{G}_3\text{P} + \text{DHAP}$$

$$\text{DHAP} \rightleftharpoons \text{G}_3\text{P}$$

$$\text{G}_3\text{P} + \text{NAD}^+ + \text{Pi} \longrightarrow 1{,}3-\text{Bisphosphoglycerate} + \text{NADH} + \text{H}^+$$

$$1{,}3-\text{Bisphosphoglycerate} + \text{ADP} \longrightarrow 3\text{-Phosphoglycerate} + \text{ATP}$$

$$2\text{-Phosphoglycerate} \longrightarrow \text{Phosphoenolpyruvate} + \text{H}_2\text{O}$$

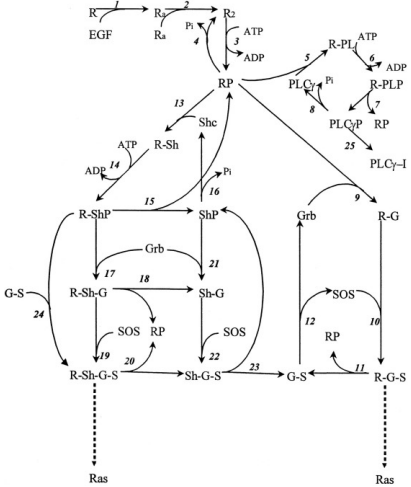
$$\text{Phosphoenolpyruvate} + \text{ADP} \longrightarrow \text{Pyruvate} + \text{ATP}$$

Overall: $\text{Glucose} + 2\,\text{ADP} + 2\,\text{Pi} + 2\,\text{NAD}^+ \longrightarrow 2\,\text{Pyruvate} + 2\,\text{ATP} + 2\,\text{NADH} + 2\,\text{H}_2\text{O} + 2\,\text{H}^+$

NOTE: Each step is an enzyme-catalyzed transformation (enzymes omitted).

NOTE: the net gain of 2 ATP arises because the pathway splits (STEP 4): each half generates 2 ATP after the cleavage step.

EGF Signaling Pathway as a Set of Reactions (Kholodenko Model)



$$\mathrm{EGF} + \mathrm{EGFR} \rightleftharpoons \mathrm{Ra}$$

$$2\,\mathrm{Ra} \longrightarrow \mathrm{R_2}$$

$$\mathrm{R_2} \longrightarrow \mathrm{RP}$$

$$\mathrm{RP} + \mathrm{Shc} \longrightarrow \mathrm{R{-}Sh}$$

$$\mathrm{R{-}Sh} + \mathrm{ATP} \longrightarrow \mathrm{R{-}ShP} + \mathrm{ADP}$$

$$\mathrm{R{-}ShP} \longrightarrow \mathrm{ShP} + \mathrm{RP}$$

etc...

Modeling CRNs:
Mathematical models of chemical kinetics

Molecules and chemical solutions

Chemical reactions describe transformation of molecules:

- A group of molecules (reactants) is transformed into another group of molecules (products)

Molecules are represented as symbols:

- A, B, C, H_2, O_2, ...

Molecules are assumed to float in a fluid medium

- called chemical solution
- the quantity of molecules is usually expressed in terms of concentrations (density)
  - $[A]$ denotes the concentration of molecule A
  - usually expressed in mol/L measure unit

Chemical reactions

Usual notation for chemical reactions:

$$\ell_1 S_1 + \ldots + \ell_\rho S_\rho \xrightarrow{k} \ell'_1 P_1 + \ldots + \ell'_\gamma P_\gamma$$

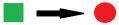
where:

- S_i, P_i are molecules (reactants and products, respectively)
- $\ell_i, \ell'_i \in \mathbb{N}$ are stoichiometric coefficients
- $k \in \mathbb{R}_{\geq 0}$ is the kinetic constant

Example: $2H_2 + O_2 \xrightarrow{5} 2H_2O$

- stoichiometric coefficients express the number of reactants/products of each type that are consumed/produced by the reaction
- the kinetic constant is a coefficient used to compute the rate of occurrence of the reaction in a chemical solution

Types of chemical reactions

Type	Reaction	Illustration
Synthesis	$\xrightarrow{k} P$	
Degradation	$S \xrightarrow{k}$	
Transformation	$S \xrightarrow{k} P$	
Binding	$S_1 + S_2 \xrightarrow{k} P$	
Unbinding	$S \xrightarrow{k} P_1 + P_2$	
Catalysis	$E + S \xrightarrow{k} E + P$	
Other	$\vdots$	$\vdots$

Types of chemical reactions

Catalysis (like other more complex types of reaction)

can be expressed in terms of binding, unbinding and transformation

Synthesis, degradation, transformation, binding and unbinding are a quite "complete" set of reaction types

Chemical reactions

Chemical reactions are often reversible

- they can occur in both directions

The usual notation for reversible chemical reaction is:

$$\ell_1 S_1 + \ldots + \ell_\rho S_\rho \underset{k_{-1}}{\overset{k}{\rightleftharpoons}} \ell'_1 P_1 + \ldots + \ell'_\gamma P_\gamma$$

where $k_{-1} \in \mathbb{R}_{\geq 0}$ is the kinetic constant of the inverse reaction transforming products P_i into reactants S_i

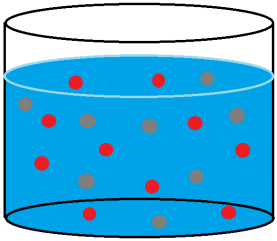
Example: $2H_2 + O_2 \underset{0.5}{\overset{5}{\rightleftharpoons}} 2H_2O$

The mass action kinetics of chemical reactions

The dynamics of chemical reactions is usually based on the following assumption:

molecules float in a (well-stirred) fluid medium (e.g. water)

Hence, they are free to move and randomly meet with each other



When a group of molecules meet, they can react

The dynamics (kinetics) of a chemical solution is modeled by the law of mass action

The mass action kinetics of chemical reactions

$$\ell_1 S_1 + \ldots + \ell_\rho S_\rho \underset{k_{-1}}{\overset{k}{\rightleftharpoons}} \ell'_1 P_1 + \ldots + \ell'_\gamma P_\gamma$$

The rate of a chemical reaction expresses the number of occurrences of such reaction in a given chemical solution in a time unit

The law of mass action is an empirical law for the computation of the rate of a chemical reaction

Definition: Law of mass action

The rate of a chemical reaction is proportional to the product of the concentrations of its reactants.

The kinetic constant is the proportionality ratio of the reaction

The mass action kinetics of chemical reactions

$$\ell_1 S_1 + \ldots + \ell_\rho S_\rho \underset{k_{-1}}{\overset{k}{\rightleftharpoons}} \ell'_1 P_1 + \ldots + \ell'_\gamma P_\gamma$$

The rate of a reaction (the left-to-right direction of the reversible reaction above) is defined as follows:

$$k[S_1]^{\ell_1} \cdots [S_\rho]^{\ell_\rho}$$

similarly, the rate of the inverse reaction (the right-to-left direction) is

$$k_{-1}[P_1]^{\ell'_1} \cdots [P_\gamma]^{\ell'_\gamma}$$

The mass action kinetics of chemical reactions

For example, the rates of

$$2H_2 + O_2 \underset{0.5}{\overset{5}{\rightleftharpoons}} 2H_2O$$

are

- $5[H_2]^2[O_2]$ for the left-to-right direction
- $0.5[H_2O]^2$ for the right-to-left direction

In a solution with the following concentrations:

- 10 mol/L of H_2
- 16 mol/L of O_2
- 30 mol/L of H_2O

rates take the following values:

- $5 \cdot 10^2 \cdot 16 = 8000$
- $0.5 \cdot 30^2 = 450$

(namely, the production of water is much faster than its decomposition)

A small comment about kinetic constants

What is the measure unit of a kinetic constant k?

It depends on the number of reactants!

- The measure unit of concentrations is mol/L
- The measure unit of the reaction rate is $mol/(L \cdot sec)$ (it is the concentration of each product produced in one unit of time)

Examples:

- $A \xrightarrow{k} B$: the rate is $r = k[A]$, that is $k = r/[A]$. So the measure unit of k is $(mol/(L \cdot sec))/(mol/L) = 1/sec$
- $A + B \xrightarrow{k} C$: the rate is $r = k[A][B]$, that is $k = r/[A][B]$. So the measure unit of k is $(mol/(L \cdot sec))/(mol/L)^2 = L/(sec * mol)$
- $2A + B \xrightarrow{k} C$: the rate is $r = k[A]^2[B]$, that is $k = r/[A]^2[B]$. So the measure unit of k is $(mol/(L \cdot sec))/(mol/L)^3 = L^2/(sec * mol^2)$

Some kinetic constants have to be changed if you change the measure unit of concentrations (e.g. μmol) or the time unit (e.g. hours)

Dynamic equilibrium

$$\ell_1 S_1 + \ldots + \ell_\rho S_\rho \underset{k_{-1}}{\overset{k}{\rightleftharpoons}} \ell'_1 P_1 + \ldots + \ell'_\gamma P_\gamma$$

A reversible reaction is said to be in dynamic equilibrium when the rates of its two directions are the same, namely:

$$k[S_1]^{\ell_1} \cdots [S_\rho]^{\ell_\rho} = k_{-1}[P_1]^{\ell'_1} \cdots [P_\gamma]^{\ell'_\gamma}$$

When the dynamic equilibrium is reached, often the chemical solution seems stable (it seems that nothing is happening inside it). Instead, reactions happen continuously, compensating each other.

It is easy to see that at the equilibrium we have:

$$\frac{k}{k_{-1}} = \frac{[P_1]^{\ell'_1} \cdots [P_\gamma]^{\ell'_\gamma}}{[S_1]^{\ell_1} \cdots [S_\rho]^{\ell_\rho}}$$

Dynamic equilibrium: simple example

As an example, let's consider:

$$A + B \underset{2.6}{\overset{0.2}{\rightleftharpoons}} C$$

with the following initial concentrations:

- $[A]_0 = 10\ mol/L$
- $[B]_0 = 15\ mol/L$
- $[C]_0 = 2\ mol/L$

the forward rate is $0.2 \cdot 10 \cdot 15 = 30$ and the backward rate is $2.6 \cdot 2 = 5.2$, hence the production of C is faster than the production of A and B.

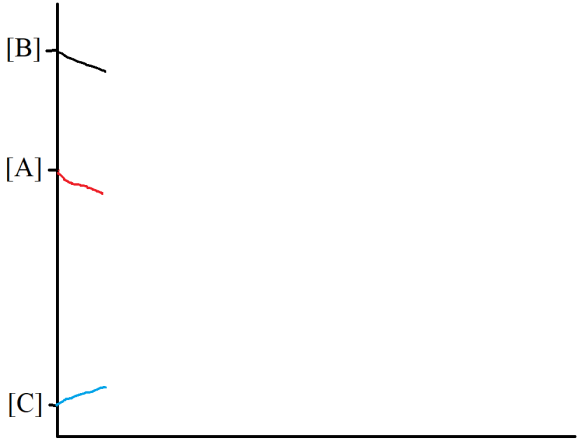
So, at the beginning the concentrations of A and B decrease, and the concentration of C increases

This causes the forward rate to decrease and the backward rate to increase, until they become equal (dynamic equilibrium), that is when

- $0.2[A][B] = 2.6[C]$

Dynamic equilibrium: simple example

We have more or less this initial dynamics:



Dynamic equilibrium: simple example

$$A + B \underset{2.6}{\overset{0.2}{\rightleftharpoons}} C$$

In order to determine the concentrations at the steady state, we can exploit the following conservation properties:

- $[A] + [C]$ is always constant (equal to $[A]_0 + [C]_0 = 12$)
- $[B] + [C]$ is always constant (equal to $[B]_0 + [C]_0 = 17$)

since each A and each B are transformed into a C, and viceversa)

So, from the formula of dynamic equilibrium, we can derive:

$$\frac{0.2}{2.6} = \frac{[C]}{[A][B]} \implies \frac{1}{13} = \frac{[C]}{(12 - [C])(17 - [C])} \implies$$

$$\implies [C]^2 - 42[C] + 204 = 0 \implies [C] \simeq 36.395 \text{ or } [C] \simeq 5.605$$

Dynamic equilibrium

Since $[A] = 12 - [C]$ and $[B] = 17 - [C]$, the two obtained values for $[C]$ give the following two candidate solutions:

- $[A] \simeq -24.395$, $[B] \simeq -7.395$ and $[C] \simeq 36.395$
- $[A] \simeq 6.395$, $[B] \simeq 11.395$ and $[C] \simeq 5.605$

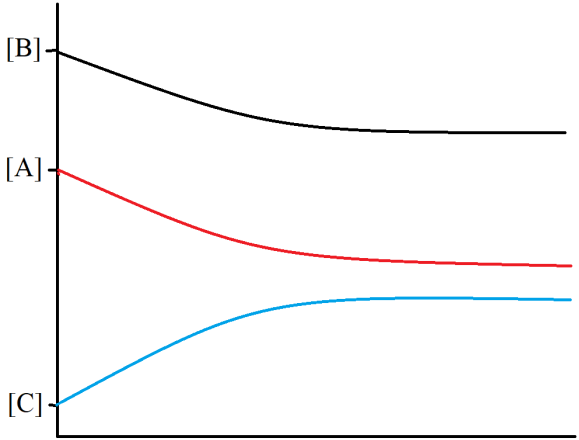
The first one is not acceptable (negative), so the second one is the steady state.

Indeed, those value make the rates of the two reactions equal:

$$0.2[A][B] = 0.2 \cdot 6.395 \cdot 11.395 \simeq 14.574$$
$$2.6[C] = 2.6 \cdot 5.605 \simeq 14.574$$

Dynamic equilibrium: simple example

We have more or less this dynamics:



From reactions to ODEs

Reaction rates can be used to construct a ODE model of the dynamics of a set of chemical reactions

The ODE model of a set of chemical reactions contains one differential equation for each molecular species

For each molecule S, an ODE is constructed by applying these two rules:

- for every reaction R in which S occurs as a reactant, its ODE contains a negative term

$$-\ell r$$

  where ℓ is the stoichiometric coefficient of reactant S in R and r is the rate of R
- for every reaction R in which S occurs as a product, its ODE contains a positive term

$$+\ell r$$

  where ℓ is the stoichiometric coefficient of product S in R and r is the rate of R

From reactions to ODEs

From these two generic reactions:

$$\ell_1 S_1 + \ldots + \ell_\rho S_\rho \underset{k_{-1}}{\overset{k}{\rightleftharpoons}} \ell'_1 P_1 + \ldots + \ell'_\gamma P_\gamma$$

we obtain the following ODEs:

$$\begin{cases} \frac{d[S_1]}{dt} = \ell_1 k_{-1} [P_1]^{\ell'_1} \cdots [P_\gamma]^{\ell'_\gamma} - \ell_1 k [S_1]^{\ell_1} \cdots [S_\rho]^{\ell_\rho} \\ \vdots \\ \frac{d[S_\rho]}{dt} = \ldots \\ \frac{d[P_1]}{dt} = \ell'_1 k [S_1]^{\ell_1} \cdots [S_\rho]^{\ell_\rho} - \ell'_1 k_{-1} [P_1]^{\ell'_1} \cdots [P_\gamma]^{\ell'_\gamma} \\ \vdots \\ \frac{d[P_\gamma]}{dt} = \ldots \end{cases}$$

From reactions to ODEs

$$2H_2 + O_2 \underset{0.5}{\overset{5}{\rightleftharpoons}} 2H_2O$$

ODEs obtained from mass action kinetics:

$$\begin{cases} \frac{d[H_2]}{dt} = 2 \cdot 0.5[H_2O]^2 - 2 \cdot 5[H_2]^2[O_2] \\ \frac{d[O_2]}{dt} = 0.5[H_2O]^2 - 5[H_2]^2[O_2] \\ \frac{d[H_2O]}{dt} = 2 \cdot 5[H_2]^2[O_2] - 2 \cdot 0.5[H_2O]^2 \end{cases}$$

$$E + S \underset{1000}{\overset{0.1}{\rightleftharpoons}} ES \underset{0.01}{\overset{0.3}{\rightleftharpoons}} E + P$$

ODEs obtained from mass action kinetics:

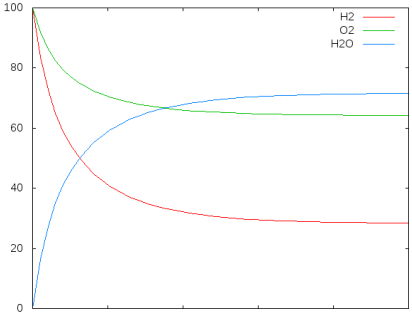
$$\begin{cases} \frac{d[E]}{dt} = 1000[ES] + 0.3[ES] - 0.1[E][S] - 0.01[E][P] \\ \frac{d[ES]}{dt} = 0.1[E][S] + 0.01[E][P] - 1000[ES] - 0.3[ES] \\ \frac{d[S]}{dt} = 1000[ES] - 0.1[E][S] \\ \frac{d[P]}{dt} = 0.3[ES] - 0.01[E][P] \end{cases}$$

From reactions to ODEs

We can now simulate the dynamics of the reactions (from given initial concentrations) by using numerical integration

$$2H_2 + O_2 \underset{0.5}{\overset{5}{\rightleftharpoons}} 2H_2O \qquad \begin{cases} \frac{d[H_2]}{dt} = 2 \cdot 0.5[H_2O]^2 - 2 \cdot 5[H_2]^2[O_2] \\ \frac{d[O_2]}{dt} = 0.5[H_2O]^2 - 5[H_2]^2[O_2] \\ \frac{d[H_2O]}{dt} = 2 \cdot 5[H_2]^2[O_2] - 2 \cdot 0.5[H_2O]^2 \end{cases}$$

with initial concentrations $[H_2]_0 = [O_2]_0 = 100$ and $[H_2O]_0 = 0$.



Numerical solution of ODEs

Sometimes, the solution of a a system of ODEs can be computed analytically, but often this is not always possibile/feasible

Very often, ODEs are studied by using numerical solvers (or numerical simulators)

- These tool solve the initial value problem (or Cauchy problem)

Definition: Initial value problem

Given an ODE $\dot{N}(t) = f(N(t))$ and an initial value N_0 such that $N(0) = N_0$, compute a function $F(t)$ that is a solution of the ODE and such that $F(0) = N_0$

The Euler method

The Euler method is the simplest numerical simulation method

It is based on the idea of discretizing the dynamics of differential equations by time steps of constant length τ.

At each step, the solution of the differential equation is approximated by its derivative computed at the beginning of the time interval of length τ

Given an ODE

$$\dot{N}(t) = f(N(t))$$

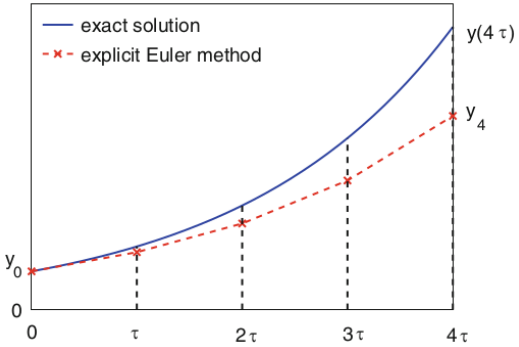
this corresponds to approximating its solution with the following recurrence relation (with $N_0 = N(0)$)

$$N_{k+1} = N_k + \tau f(N_k)$$

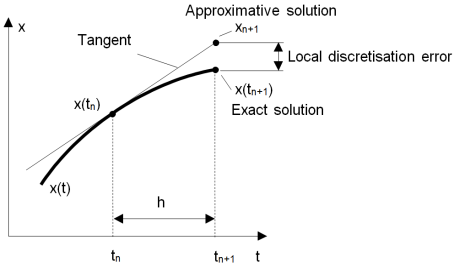
where N_k approximates $N(k\tau)$

The Euler method

Example of execution of the Euler method:



The Euler method: errors



Each step of the Euler method gives rise to an error

- Local discretization error (or local truncation error) $|N(\tau) - N_1|$
- it is in the order $O(\tau^2)$

Errors accumulate: after k steps, namely at time $t = k\tau$, we have

- Global discretization error (or global truncation error) $|N(k\tau) - N_k|$
- it is in the order $O(k\tau^2) = O(\tau)$, since $k\tau = t$ is constant

Other "explicit" methods

A linear ($O(\tau)$) global discretization error often imply that a very small discretization step τ has to be used

- the computation becomes very slow (many steps)

Other methods have a global discretization error of a higher order (e.g. $O(\tau^p)$ for some p) which is better as long as $\tau \to 0$ (hence, it is smaller than one)

A few examples of such methods:

- Runge-Kutta methods: $p = 2$ in their original formulation, but can be higher
- Multistep methods (e.g. Adams methods): extrapolate the value of the next step from the values of the previous k steps. $p \approx k$

Other "explicit" methods

State-of-art methods can also:

- self-determine the step size τ based on thresholds on local and global discretization errors
- dynamically adjust the step size τ during their execution (e.g. Adaptive Runge-Kutta)

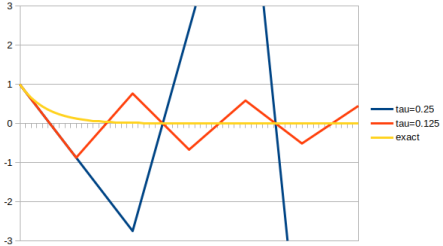
Instability and stiff systems

In some cases, explicit methods may become unstable, unless a very small step size τ is used.

Example:

$$\dot{N}(t) = -15N(t)$$

with $N(0) = 1$



Euler method with $\tau = 0.25$ and $\tau = 0.125$
compared with the exact solution

Instability and stiff systems

This kind of problematic systems are called stiff systems

There is no precise definition of stiffness

- Intuitively, the system contains some very fast term which causes very small step sizes to be used with explicit methods
- Sometimes, the combination of fast terms with slow terms in a system of ODEs make explicit methods unstable although the shape of the solution is smooth

Implicit Euler method

Implicit methods are often better suited for stiff systems.

Implicit variant of the Euler method:

At each step, the solution of the differential equation is approximated by its derivative computed at the end of the time interval of length τ

Given an ODE

$$\dot{N}(t) = f(N(t))$$

this corresponds to approximating its solution with the following equation (with $N_0 = N(0)$)

$$N_{k+1} = N_k + \tau f(N_{k+1})$$

where N_k approximates $N(k\tau)$

Implicit Euler method

$$N_{k+1} = N_k + \tau f(N_{k+1})$$

In this case the value of N_{k+1} is not explicitly expressed in terms of N_k, but it is implicitly espressed by means of an equation which could be difficult to solve

Hence, the computation of the single step requires more effort in implicit methods

But, often the local discretization error is smaller

- greater values of τ can be used
- with stiff systems it is often convenient to pay the extra time for the computation of each step

Other "implicit" methods

A few examples of implicit methods:

- Implicit Runge-Kutta methods
- Multistep methods (e.g. BDF methods)

Implementations of implicit methods often require the modeler to provide the Jacobian matrix (partial derivatives) of the function f

Also in these cases there are variants that can self-determine and dynamically adjust the step size τ according to threshold on the local and global discretizaton errors.

There exist methods that are able to automatically switch from explicit to implicit modes (and vice-versa)

- LSODE: switches between Adams and BDF
- CVODE: switches between Adams and BDF

Numerical integration of ODEs in Python

Python stack for ODEs and CRNs

- NumPy: vectorized arrays and linear algebra.
- SciPy.integrate: ODE solvers (`solve_ivp`, `odeint`).
- Matplotlib: plotting trajectories and phase portraits.

SciPy: solve_ivp (adaptive)

- RK45: good default for non-stiff systems (Runge-Kutta method, 5th-order, adaptive step).
- Radau, BDF: implicit, suitable for stiff systems.
- rtol, atol control error; t_eval sets output times.

```
from scipy.integrate import solve_ivp
import numpy as np

# A + B <-> C (mass-action)
def f(t, y):
    A, B, C = y
    kf, kr = 0.2, 2.6
    r_f = kf*A*B
    r_r = kr*C
    return [-r_f + r_r, -r_f + r_r, +r_f - r_r]

y0 = [10.0, 15.0, 2.0]
t_eval = np.linspace(0.0, 50.0, 400)
sol = solve_ivp(f, (0.0, 50.0), y0,
                method="RK45", rtol=1e-6, atol=1e-9,
                t_eval=t_eval)
```

Examples and exercises on Colab!

See examples and exercises on Colab:

https://colab.research.google.com/drive/1_hR3_EFB9BuUTfmFOqKSB7jZcOdb0TQi?usp=sharing